

Equilibrium Imitation and Growth

Perla & Tonetti (JPE 2014)

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Motivation & Literature

- **Frontier Innovation Models:** Growth driven by expanding the technological edge via R&D (e.g., Romer 1990; Aghion & Howitt 1992).
- **Empirical Reality:** Vast majority of firms operate far below the frontier; mass is concentrated in the lower tail (Gabaix 2009).
- **Knowledge Diffusion:** Shifting the mass upward via random encounters and learning (e.g., Alvarez et al. 2008; Lucas & Moll 2014).
- **This Paper:** Endogenizes the search and adoption choices of the **least productive** laggard firms to generate self-sustaining growth.

Environment: Consumers

Representative Consumer

- CRRA preferences over aggregate consumption Y_t :

$$u(Y_t) = \frac{Y_t^{1-\gamma}}{1-\gamma}$$

- Euler equation defines the interest rate r_t :

$$\frac{1}{1+r_t} = \beta \frac{u'(Y_{t+1})}{u'(Y_t)}$$

The Search Problem

The option value of search depends on the shape of the productivity distribution.

Firm's Bellman Equation

$$V_t(z) = \max \left\{ z + \frac{1}{1+r_t} V_{t+1}(z), \frac{1}{1+r_t} \int_{m_{t+1}}^{\infty} \frac{V_{t+1}(z') f_t(z')}{1 - F_t(m_{t+1})} dz' \right\}$$

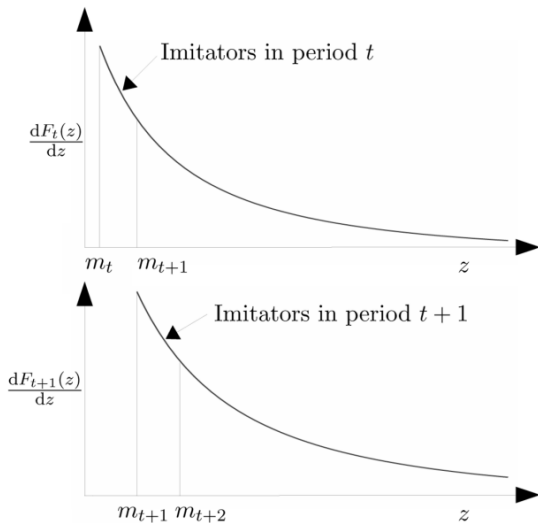
- Solved via a reservation productivity threshold $\hat{z}_t$.
- Firms search if and only if $z \leq \hat{z}_t$.

Law of Motion

- **Rational Expectations Equilibrium:** Individual thresholds perfectly track the shifting bottom of the support: $\hat{z}_t = m_{t+1}$.
- **"Meet-and-Copy" Technology:** Searching firms clear out the bottom and sample from active producers.
- This results in a clean **sequential truncation** of the density:

$$f_{t+1}(z) = \frac{f_t(z)}{1 - F_t(m_{t+1})}$$

Visualizing the Productivity Shift



The "Tractability Trick" – Uniqueness of Pareto

- **Scale Invariance:** For a Balanced Growth Path (BGP), the normalized distribution shape must remain constant.
- Combining scale invariance with the law of motion yields an ordinary differential equation (ODE) in density space.
- **Result:** The **Pareto Distribution** is the unique functional form satisfying a BGP:

$$F_0(z) = 1 - \left(\frac{m_0}{z}\right)^\alpha, \quad \alpha > 1$$

BGP Competitive Equilibrium Solution

By guessing and verifying that $m_{t+1} = gm_t$ and $V_t(gm_t) = m_t W$:

Analytical Growth Factor (g_{ce})

$$g_{ce} = \left(\beta \frac{\alpha}{\alpha - 1} \right)^{\frac{1}{\gamma - 1 + \alpha}}$$

- $g_{ce} > 1$ represents sustained endogenous growth.
- Growth is generated purely as an externality from risk-taking at the bottom of the distribution.

BGP Comparative Statics

$$g_{ce} = \left(\beta \frac{\alpha}{\alpha - 1} \right)^{\frac{1}{\gamma - 1 + \alpha}}$$

- $\frac{dg}{d\beta} > 0$: Patient agents value the future more $\rightarrow$ search increases.
- $\frac{dg}{d\gamma} < 0$: Higher risk aversion reduces willingness to substitute consumption over time.
- $\frac{dg}{d\alpha} < 0$: Lower $\alpha \rightarrow$ **higher inequality** (fatter right tail).
- Fatter tails expand opportunities, raise the option value of search, and fuel growth.

Non-BGP Dynamics & Asymptotic Growth

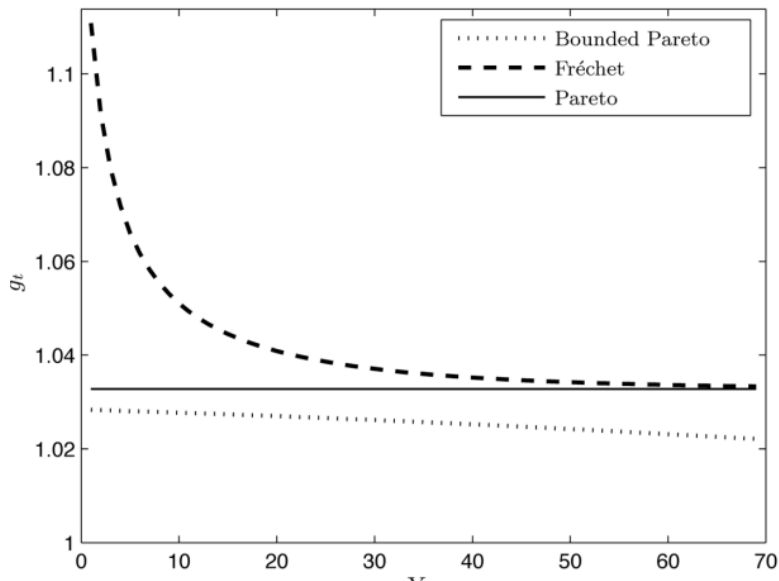
What if the initial distribution $F_0(z)$ is not Pareto?

- **Power Laws (e.g., Fréchet):** Possess a Pareto right tail. The non-Pareto body washes out over time, and growth asymptotically converges to $g(\alpha)$.
- **Non-Power Laws (e.g., Lognormal, Exponential, Bounded):** Tail expectations collapse relative to the support limit:

$$\lim_{m \rightarrow \infty} \frac{\mathbb{E}[z|z > m]}{m} = 1$$

Returns to search die out, and growth stops ($\lim_{t \rightarrow \infty} g_t = 1$).

Transition Dynamics Off the BGP



The Social Planner Problem

- **The Friction:** Atomistic firms do not internalize how searching upgrades the search pool for future generations.

Planner's First-Best Growth Rate

$$g_{fb} = \left(\beta \frac{\alpha}{\alpha - 1} \right)^{\frac{1}{\gamma - 1}}$$

- Competitive equilibrium has a growth penalty (α) in the exponent:

$$g_{ce} = \left(\beta \frac{\alpha}{\alpha - 1} \right)^{\frac{1}{\gamma - 1 + \alpha}}$$

- Consequently, $g_{fb} > g_{ce}$ always.

Policy: Subsidies and Taxes

Can a constrained planner restore the first-best using linear instruments?

- **Tension:** Subsidizing search (ζ) shifts mass upward, but balancing the budget requires taxing production (τ), which dampens search incentives.

Constrained Planner's Setup

$$\begin{aligned} \max_{\tau, \zeta} \quad & U(F) \\ \text{s.t.} \quad & \tau \int_{gm(F)}^{\infty} z \, dF(z) = \zeta \int_{m(F)}^{gm(F)} z \, dF(z) \end{aligned}$$

- **Result:** A constant, feasible policy sequence (τ^*, ζ^*) perfectly implements the unconstrained first-best g_{fb} .

Extension: Unconditional Draws

What if search is unconditional

- Firms sample from the entire distribution, opening the door to low draws.
- Firms can reject draws beneath their current level ($\max\{z', z\}$).
- **Congestion Effect:** Drawing below the search threshold forces firms into perpetual search.
- Quantitatively slows down growth, but the qualitative mechanism remains intact.

Conclusion & Discussion

- **Takeaway:** An elegant, analytically tractable engine for endogenous growth driven entirely by laggard firms catching up.
- **Inequality Paradox:** Greater productivity dispersion expands option values, generating faster aggregate growth paths.